

Topological 0-Matrix _(Substrate) Mechanics (TSM)

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Chapter 0:

Axiomatic Ontology of the 0-Matrix — The Mechanical Engine of Reality

This monograph presents the foundations of Topological Substrate Mechanics (TSM) — a new, mechanistic paradigm for describing reality that enables a coherent reconstruction of the key results of General Relativity and the Standard Model. The fundamental premise of TSM is not an attempt at a formal synthesis of both theories at their current phenomenological level. Instead, the formalism descends one level lower — to the sub-Planckian structural mechanics of the microworld — and from the geometric laws operative there derives macroscopic phenomena such as time dilation and the constancy of the speed of light.

Within the TSM paradigm, velocities exceeding the speed of light are a natural phenomenon, realised, however, in the fourth spatial dimension. In our three-dimensional world, the speed of the electromagnetic wave defines solely the upper bound on the propagation of shear disturbances in a jammed medium. At the deepest microscopic level, the primordial 0-Particles oscillate at velocities far exceeding the macroscopic speed of light. These fundamental micro-vibrations do not, however, imply the passage of time in the physical sense — their sole function is to generate the internal stresses, pressure, and elasticity of the entire substrate. Classical relativistic time emerges only at a higher organisational level, as an emergent measure of the succession of repeatable states of organised matter permanently trapped within the geometry of our 3-brane.

Furthermore, TSM radically simplifies the architecture of the Standard Model. It eliminates the need to postulate a multitude of independent elementary particles and fundamental direct interactions. We demonstrate that all observable manifestations of matter and forces reduce to geometrically differentiated deformations, entanglements, and excitation states of a single universal medium. TSM unifies the achievements of theoretical physics to date, translating the complex apparatus of the mathematical continuum into the rigorous and unambiguous language of discrete lattice mechanics.

0.1. Rejection of the Unphysical Void

One of the most serious conceptual crises in contemporary fundamental physics concerns the ontological status of the vacuum. Classical mechanical intuition defines a wave unambiguously as a dynamic disturbance of a specific material medium. Classical physics accepts this fact unreservedly at the macroscopic scale; yet in the domains of electrodynamics and field theory, the prevailing dogma holds that electromagnetic or gravitational waves are capable of transporting energy and momentum through an absolute, unphysical nothingness. TSM categorically rejects this logical inconsistency.

We introduce the first axiom of the theory: space is not empty. It is entirely filled with fundamental, discrete elements — 0-Particles. Their linear dimensions lie deep within the sub-Planckian scale, rendering them absolutely undetectable by current empirical methods. They constitute the homogeneous, ultra-efficient substrate of reality — the titular 0-Matrix.

These particles are devoid of electric charges, internal structures, or any fields of their own. As elementary, indivisible geometric units, they are characterised by identical inertial mass, understood solely as an invariant parameter of resistance to any change in their oscillatory state. This mass — uniform and identical for every 0-Particle — does not constitute a direct source of the gravitational field: as a property of a homogeneous background it generates no gradient, and it is the gradient, not the mere filling of space with mass, that is the essence of gravitational interaction.

The source of gravitation in TSM is instead the permanent deformation of the substrate within the 3-brane — that is, the topological knot constituting an elementary particle. Such a deformation is not locally neutral with respect to its surroundings: the particle, by virtue of its bending into the fourth spatial dimension, exerts a persistent pressure on that dimension — it places it in tension. This tension of the fourth dimension, generated by the deformation, is the physical content of what we observe as gravitation. Consequently, the quantity referred to in the macroworld as “mass” is not identical to the inertial mass of an individual background 0-Particle; it is instead a measure of the depth of that topological bending: the stronger the deformation of the brane into the fourth dimension, the greater the pressure exerted on that dimension, and hence the greater the gravitational mass observed within our three-dimensional world.

Each 0-Particle possesses a strictly defined, non-zero proper volume. We thereby exclude from the mathematical formalism the abstract, unphysical concept of a material point of zero dimension, which in classical field theories is the direct source of energy infinities. Since the light-speed barrier and the Lorentz transformations in TSM are exclusively a secondary, wave-like property of the 3-brane structure, the primordial 4D mechanics at the sub-Planckian level is absolute and non-relativistic in character. This enables the realisation of momentum and kinetic energy at velocities far exceeding the speed of light, without generating relativistic singularities for which there is no structural basis at this scale.

In this picture every energetic state of the substrate reduces to pure collision mechanics — kinetic energy arising from the incessant, rapid oscillations of the spherical representations of these particles. These primordial micro-oscillations neither measure nor define the passage of time; they constitute solely the mechanical energy source of the background, responsible for generating the global pressure and local stress gradients within the 0-Matrix. Everything we observe as macroscopic physical phenomena emerges as an echo of the geometry of these stresses.

0.2. Elasticity from Pure Kinematics and the Concept of Virtual Time

Orthodox physics interprets the elasticity of media by postulating inter-particle forces — potentials of attraction and repulsion. Within the TSM formalism we reject this description as methodologically flawed, since it introduces the concept of force within the structure before explaining the very nature and origin of the interaction. We introduce no additional force fields. All macroscopic elasticity of the Substrate is derived solely from the spatial collision dynamics of rigid 0-Particles.

Since relativistic time is an emergent quantity and does not exist at the lowest structural level of the 0-Matrix, we introduce an auxiliary operational instrument for describing the kinematics of the primordial components prior to the emergence of the macroscopic flow of phenomena: the **virtual second**. This constitutes an elementary sub-temporal interval determined solely by the microscopic dynamics of a single 0-Particle — one virtual second corresponds to exactly one oscillation cycle (one collision) of the particle against the boundaries of its fundamental configurational cell. Such an axiomatisation permits rigorous use of the concepts of frequency and velocity at the micro-scale, without committing the error of circular reasoning in the subsequent derivation of emergent time.

The introduction of virtual time makes it possible to formulate a key thesis: primordial 0-Particles are exclusively indivisible, rigid geometric objects of fixed radius and inertial mass. The apparent “interaction sphere” of critical radius is a phenomenon purely kinetic in character, arising from the extreme velocity of the micro-oscillations.

Decisive here is the geometry of the configurational cell: its walls are not formed by a smooth, empty cavity, but by the convex interaction spheres of neighbouring 0-Particles. Motion within a region bounded by convex scattering obstacles constitutes a system known in the theory of dynamical systems as the **Sinai billiard** (Lorentz gas). In contrast to a billiard with smooth, focusing walls (e.g. an ideal hollow sphere, where the motion is integrable and leaves an unfilled central core), a scattering billiard of this type is not only ergodic but exhibits the stronger property of **mixing**: regardless of the initial condition, the spatial distribution of the particle’s presence converges exponentially rapidly to a uniform equilibrium distribution over the entire accessible volume of the cell.

Since the oscillation frequency of a 0-Particle, expressed in virtual seconds, exceeds by tens of orders of magnitude any macroscopically distinguishable unit of time, the time required

to attain this mixing is, from the perspective of emergent relativistic time, practically zero. The particle therefore does not “traverse” its interaction sphere in any observable time — it fills it with its physical presence almost instantaneously and continuously. From an external perspective this creates a dynamic, impenetrable shield with spherical symmetry.

To formalise this mechanism and connect it to the geometric packing limits of the substrate, consider a 0-Particle confined within a local cage defined by its neighbours. The frequency with which the particle strikes the geometric boundaries of its cell depends directly on the free space remaining between the physical cores. When the substrate is compressed by external macroscopic stresses, the distance between the centres of neighbouring particles decreases, reducing the mean free path to a limit determined by the sum of the radii of the rigid cores. In this limiting regime the free volume tends to zero, implying an asymptotic, hyperbolic increase in collision frequency.

Since in the elastodynamic picture force is a measure of momentum transfer per unit time, the incessant reflections of the particle generate an effective, time-averaged repulsive force acting on the cell walls. This force tends to infinity as one attempts to compress the cage to the limit of the physical particle cores, creating the complete illusion of a nonlinear potential force field with an infinite repulsive barrier. For the structure of the substrate, this sharp increase in collision density signifies a powerful spike in local pressure. This creates a highly elastic structure that resists compression — not through hidden fields, but through pure geometry and momentum transfer. Stress gradients are an emergent property of the increased collision density in confined space.

Volumetric compression cannot, however, proceed indefinitely. Before the system reaches the unphysical state of ideal contact between core boundaries, the local stress tensor exceeds a critical threshold. This triggers the Critical Displacement Clause — the substrate undergoes plastic flow, which relaxes the excessive collision density through network reconfiguration. This mechanism defines a hard limit for stable dynamic packing: the maximum packing threshold of spheres does not represent direct, static contact of the bare, indivisible core boundaries. It is the state of maximum effective-sphere compaction at which the asymptotic increase in repulsive force prevents any further reduction in the distance between particle centres in the elastic regime.

0.3. The Critical Regime of the Substrate, the Asymptotic Force Barrier, and the Mechanism of Plastic Flow

If the fundamental space were infinite, the primordial kinetic activity of the 0-Particles would disperse irreversibly, and mutual interactions would cease in a process of eternal expansion. For reality to exist as a stable structure capable of sustaining deformations, it must satisfy a condition of geometric finiteness.

Our fundamental space is a four-dimensional, finite, yet geometrically closed manifold — the structural analogue of a hypersphere. The absence of physical boundaries or external walls

does not imply infinity — the substrate closes in on itself, constituting a closed environment completely filled with 0-Particles. From its initial state it resides in a state of global jamming, where the global geometric packing fraction exceeds the critical value.

This means that the 0-Particles are permanently confined within their configurational cells and possess no translational freedom, i.e. no macroscopic flow. The ground state is therefore not a chaotic free gas, but a rigid, elastic topological glass in 4D, capable of sustaining and generating powerful stress gradients in all four spatial dimensions. The only remnants of the primordial dynamics are local fluctuations — minor oscillations about the geometric interaction centres — which do not violate the global topology of the medium.

Since the 0-Particles are geometrically constrained with no possibility of escape, they are structurally compelled to exert permanent pressure on their neighbours. This persistent, kinematic jammed state means the substrate is never relaxed — it remains in a state of perpetual, background tension. In this picture the Universe is not secondarily filled with energy; it itself is that kinetic energy, locked and trapped within the finite, closed geometry of the substrate.

0.3.1. Asymptotic Prohibition of Physical Contact

A fundamental condition for preserving the dynamic, oscillatory nature of the 0-Matrix is the prohibition of the permanent, static locking of its constituent elements. Although 0-Particles are characterised by an incompressible, rigid core of finite volume, direct contact between their physical boundaries is impossible. This follows from the nature of the sub-Planckian interaction sphere. The repulsive force between two neighbouring 0-Particles grows nonlinearly as the distance between their geometric centres decreases. As this distance approaches the critical threshold, the force tends asymptotically to infinity, with the threshold distance strictly greater than the sum of the radii of the rigid cores.

By virtue of this asymptotic force barrier, the inter-particle distance — even under conditions of extreme topological compression — remains always at a minimum, but strictly greater than zero. A consequence of this mechanism is the complete exclusion of physical contact between the core spheres, which prevents their permanent, static locking and guarantees that the substrate retains its state of continuous micro-oscillations under all conditions. This prohibition applies absolutely and locally — for each pair of neighbouring 0-Particles individually — regardless of the global stress regime prevailing in the substrate. As shown later in this chapter, even in states of plastic network reconstruction this barrier is never breached; only the topology of the neighbourhood changes, not the rule of minimal distance itself.

0.3.2. The Critical Displacement Clause

When an extreme external pressure or a local topological twist acts on a given region of the 0-Matrix, inter-particle distances decrease drastically, inducing a gigantic increase in local

energy density and shear stresses. In this regime the network reaches the state of maximum permissible dynamic packing. If the local shear stress exceeds the critical plasticity threshold of the substrate, the Critical Displacement Clause is activated.

Since the asymptotic force barrier admits no exceptions, no single 0-Particle is capable of independently “jumping” into a neighbouring cell — this would require momentary approach to the particles forming its boundary to within the critical distance, which is categorically forbidden. The Critical Displacement Clause therefore describes an inherently collective phenomenon: a coordinated reconstruction of the neighbourhood topology within an entire local cluster of 0-Particles, rather than the motion of a single particle through another’s barrier.

This mechanism consists in circumventing the barrier within the network, not in breaching it. A group of neighbouring 0-Particles executes a coupled, synchronous shift of their equilibrium positions, during which the distance between no pair of particles at any moment falls below the critical distance. A necessary condition for such a shift is the existence, in the immediate neighbourhood, of a complementary region of local rarefaction — a fragment of the substrate that momentarily “yields”, absorbing the excess packing from the critical region. The excess stress does not therefore vanish at a single point, but is transmitted further into the substrate, initiating a cascade of local rearrangements, each of which individually respects the asymptotic contact prohibition.

The net effect of this cascade is that individual 0-Particles acquire new neighbours — the structure of their configurational cages changes — without the minimal inter-particle distance being violated at any moment in the process. This is strictly analogous to the reconstruction of the contact network in jammed granular media or colloidal glasses: the system changes its local structural connectivity not by elements penetrating through one another, but through a cooperative, wave-like shift of the entire neighbourhood.

Inside the critical zone: the cluster of 0-Particles encompassing the overloaded region and the adjacent compensatory region acts as a coherent carrier of plastic deformation (a soliton) — none of the participating particles entering physical contact with the cores of the others.

On leaving the critical zone: at the moment the rearrangement wave moves away from the original stress source and the local shear stress falls below the plasticity threshold, the participating 0-Particles immediately and fully recover their original, isotropic interaction spheres — this time centred about their newly established equilibrium positions.

Upon recovery of the interaction spheres, the capacity for further translation is immediately blocked. Each 0-Particle is once again trapped within a newly formed potential well, returning to its fundamental regime: rapid, local micro-oscillations about the new equilibrium position. The energy of the original disturbance is thereby dissipated — not through breach of the force barrier, but through its topological circumvention — and the substrate at that point returns to its elastically frozen state. In the low- and medium-energy regime the translational freedom of a single particle unsynchronised with the rest of the network is zero, which guarantees the structural stability of the substrate and of the topological knots embedded within it.

0.4. The Relaxation State: Self-Regulation of the 0-Matrix

Every point of the 0-Matrix tends toward a relaxation state — a configuration in which the contact pressure from every direction is perfectly balanced by that of the neighbouring 0-Particles. At the micro-scale this means that a local compression of the substrate immediately induces a sharp increase in collision intensity, which the system manifests as a local excess of stress.

This phenomenon is strictly analogous to stress relaxation in a jammed granular medium or an overloaded solid: since the particles possess no translational freedom and cannot physically flow away, the excess kinetic energy is inevitably transmitted further, from cell to cell, through a cascade of contact collisions. The system does not allow excess energy to remain in one location — it immediately disperses it in wave form into the surrounding regions, tending toward the equalisation of configurational pressure throughout the entire substrate.

This mechanism ensures that no local deformation can persist in the low-energy regime. If any external agent disturbs this arrangement, an elastic wave will propagate through the system until the medium returns to equilibrium.

This constant tendency toward stress equalisation we define as the relaxation of the 0-Matrix. The dominant source of the observed cosmological redshift is, however, not this mechanism, but a distinct phenomenon: condensation of the 0-Matrix induced by the gravitational densification of matter, analogous to classical gravitational redshift but extended to large-scale condensation gradients. This mechanism is developed further in the monograph.

The relaxation described in this section is, by contrast, a local and secondary phenomenon. When a sudden, point-like disturbance strikes a given region of the substrate — for example, as the result of a supernova explosion or some other energetic astrophysical event — the 0-Matrix is agitated like the surface of water after an impact. Just as an aqueous medium tends to dampen its own waves, the 0-Matrix tends to disperse that disturbance back toward equilibrium. Light traversing such a locally turbulent, not yet relaxed region undergoes additional, minor energy loss — a “fatigue” effect spatially and temporally confined to the vicinity of the source of the disturbance, and not spread uniformly across the entire path of the photon through the cosmos. This mechanism is therefore not a competing explanation to metric expansion for the cosmological redshifting of spectra at the global scale — it constitutes only a local, additive contribution in the neighbourhood of high-energy disturbance sources.

0.5. The Birth of the 3-Brane (The Great Event and the Topological Defect)

The question of how an ordered three-dimensional world emerges from a four-dimensional, jammed medium is resolved by the mechanics of topological defects. We posit that lying beyond the strict descriptive scope of TSM is a primordial, powerful disturbance — the **Great Event** — analogous in its limiting role to the initial singularity in the standard cosmological model. This event acted upon the rigid, four-dimensional 0-Matrix, inducing two coupled effects:

1. **Topological twisting:** In a certain region the substrate underwent a limiting plastic deformation. A permanent topological defect was formed in the shape of a three-dimensional hypersurface — the **3-brane**. The boundary between the twisted and untwisted portions of the medium has, by definition, a dimension one less than that of the parent space, which naturally accounts for the three-dimensionality of the observable Universe. We provisionally assume that the brane inherits the elasticity of the parent 4D substrate and is capable of transmitting transverse waves — this is not a logical necessity, but a modelling simplification to which we return below. Crucially, the macroscopic defects formed upon it are topologically anchored in this primordial twist; their departure from the 3-brane into the free 4D dimension would require energy equal to the rupture of the continuity of the entire continuum, which permanently confines all known matter within this three-dimensional world.
2. **Mega-fluctuations of pressure:** Within the newly formed 3-brane the primordial disturbance induced extreme, local stress spikes. Wherever the stress tensor exceeded the plasticity threshold and the jamming barrier, permanent, nonlinear knots were formed — four-dimensional topological nodes. These constitute the first stable particles of matter.

The 4D region beyond the 3-brane space remains a rigid, undisturbed topological glass. Since it is perfectly elastic and remains in a state of static equilibrium, it generates no additional dynamic forces acting on the brane — all observable physical dynamics of the Universe unfolds within the hypersurface defect itself.

The assumption of full, isotropic inheritance of elasticity cannot today be directly verified — it lies beyond the reach of current measurement methods. The only available route to an indirect assessment remains the observation of consequences: were the process of topological twisting to have left a residual elastic anisotropy in the brane, this should manifest as a preferred direction for the formation of large-scale structures on the brane — even though in the scenario of full, isotropic inheritance no direction should be preferred. It is worth noting that large-scale directional anomalies of this kind have long been the subject of discussion in observational cosmology — among them the so-called “axis of evil”, hemispherical asymmetry, and the alignment of low-order multipoles in the cosmic microwave background. TSM does not determine at this point whether these phenomena are related to the twist of the primordial substrate; it merely notes that the assumption of inherited brane elasticity is in principle falsifiable through observations of this type, even if their ultimate interpretation lies beyond the scope of this chapter.

0.6. The Axiom of Physical Finiteness: Categorical Exclusion of Singularities and “Black Holes”

At this point we formulate a fundamental logical postulate that constitutes the deepest line of demarcation between TSM and the geometric heritage of General Relativity and standard quantum mechanics.

TSM introduces the **Axiom of Physical Finiteness**: in the real world, infinities do not exist. Mathematical concepts in which any physical parameter — such as density, pressure, temperature, or spacetime curvature — tends to infinity are exclusively artefacts of imperfect, continuum-like mathematical models, devoid of a rigorous mechanistic foundation.

The consequence of the non-zero proper volume of the 0-Particles and the existence of a geometric limit to their packing is a categorical prohibition on the collapse of any region of space to a dimensionless point. Since 0-Particles possess hard, incompressible kinetic cores, no force in the Universe — including the most extreme gravitational pressure — is capable of compressing matter below its geometric structural limit. For this reason the TSM formalism definitively and unambiguously excludes the existence of gravitational singularities.

In this way the formalism presented here closes a complete description of the transition from the timeless sub-Planckian lattice to the relativistic macroworld, ultimately overturning the dogma of empty, geometric spacetime in favour of a coherent elasto-mechanics of the 0-Matrix. This reorientation leads to a fundamental breakthrough in astrophysics: it entirely eliminates the concept of a destructive singularity and excludes the existence of black holes, replacing these mathematical fictions with real, high-energy condensed structures — ultra-dense plasmoids. TSM precisely defines the mechanics of these objects, demonstrating that it is precisely these ultra-dense plasmoids that constitute the real, powerful engines driving the evolution of galaxies. Their direct, tangible manifestation is the giant relativistic jets observed today, which have hitherto been mistakenly attributed to the gravitational collapse of matter.

Current radio-interferometric imaging of compact structures at the centres of galaxies (EHT, M87, *Sgr A*) is not in conflict with the TSM plasmoid model. The observed shadow is not an image of a singularity or an event horizon — it is an image of a photon photosphere, which any sufficiently compact massive object generates regardless of its internal structure. The qualitative observational signature of a plasmoid and a black hole in the GR sense are indistinguishable at this level; their quantitative differences — in shadow size, edge-emission profile, and polarisation structure — constitute specific, falsifiable predictions of TSM, which we develop further in the monograph.

0.7. The Creation of the First Distortions: An Alternative to the Big Bang

Within the structure of the theoretical model we must clearly delineate the boundary of knowledge. The question of the primordial nature of the Great Event — what caused the initial twist of the 0-Matrix and the mega-fluctuations of pressure — remains an open question. Inquiry into the primordial cause lies beyond the strict scope of physical theory; it constitutes a domain of philosophical speculation rather than empirically verifiable mechanics. TSM begins operationally from the moment such a disturbance has already been set in motion within the medium.

We can, however, on the basis of the Axiom of Physical Finiteness and the picture of a primordially jammed substrate, formulate a rigorous alternative to the classical concept of

the “Big Bang”. Since the phase of an initial singularity is geometrically impossible, the TSM model rejects the one-off, unphysical act of the creation of spacetime and matter from a point of zero volume.

Instead, the model introduces a mechanically rational scenario: within our closed, four-dimensional manifold, a powerful local disturbance — a pressure impulse — occurred, which twisted a portion of the jammed substrate, forming the 3-brane, and upon it, through mega-fluctuations, massively generated the first topological nodes. Furthermore, the theory does not exclude the possibility that similar processes continue to occur in distant, unobservable regions of the brane, perpetually creating new regions of organised matter. It is in this manner — through the local surpassing of the plasticity thresholds of the substrate — that the first concentrations of matter in the Universe arose, and may continue to arise.